MULTIDISCIPLINARY SCIENCE JOURNAL

RESEARCH ARTICLE
Published Online: February 24, 2025
https://doi.org/10.31893/multiscience.2025389

Examining the role of normative influences in sustainable clothing disposal behaviors

Maghfira Rizky Maulani[a] | **Indah Fatmawati**[a] ✉ | **Nuryakin**[a]

[a]Universitas Muhammadiyah Yogyakarta, Indonesia.

Abstract Over the past ten years, there has been a growing awareness on a global scale regarding environmental concerns, particularly in relation to the environment's sustainability and waste management. People are encouraged to adopt more sustainable consumption practices, including clothing disposal. This research field examines consumers' complex post-consumption practices, including how they choose sustainable disposal methods, consisting of philanthropic clothing disposal (donating, giving) and economical clothing disposal (reselling), especially in young adults consumers in Indonesia. Purposive sampling was used, and 251 respondents were found. Data were analyzed using SPSS 23 and AMOS SEM 29 after being collected through Google Forms. This study's findings suggest that social norms significantly influence donation and resell behavior but do not affect giving. Conversely, subjective norms substantially impact all sustainable disposal behaviors, including reselling, donating, and giving. In contrast to environmental awareness, it has been demonstrated to have no significant effect on donating, giving, or reselling. Our hypothesis about environmental awareness and norms in sustainable apparel redistribution methods needs to be revised. The subjective norm is the biggest aspect influencing all sustainable behaviors and goes against environmental awareness. Despite this, social norms do not affect giving. This suggests customers are more willing to participate in sustainable disposal practices through philanthropic disposal (donating, giving) or economic disposal (reselling) when motivated or intervened by people they respect, particularly young adults.

Keywords: social norms, subjective norm, environmental awareness, donating, giving, resell

1. Introduction

The fashion industry in the marketing 4.0 era is increasing, especially in the apparel industry. Many manufacturers make cheap and trendy clothes but live only briefly. The clothes that circulate on the market today usually follow only the trend that consumers are hyping. Meanwhile, trends are constantly changing, so clothes that only follow the trend will give rise to the desire to immediately throw away and replace them with new, trendier clothes. This is undoubtedly beneficial from a business perspective but detrimental if we look at it from an environmental health perspective. If left unchecked, the accumulation of clothing waste will increase daily and pollute the water, soil, and air we breathe. In today's environment, consumer demand for new clothing and the fast fashion industry have led to overconsumption and environmentally unfriendly disposal practices. (Cudlínová et al., 2022).

At landfills, carbon dioxide and methane, two heavy greenhouse gases, are released (Lamb et al., 2014). Hoarding is not more detrimental to the environment or public health than burning textiles. Improper management of fashion trash can lead to ethical, societal, and environmental problems (Brewer, 2019). Several initiatives have been suggested to address these issues and promote more sustainable practices in the apparel industry, including interventions aimed at consumers, suppliers, buyers, and retailers (Harris et al., 2016).

In the last decade, global attention to environmental issues has been increasing, especially with respect to waste management and environmental sustainability. Various campaigns and social movements encourage people to adopt more sustainable consumption patterns, including clothing disposal. The current increase in clothing production and consumption requires sustainable clothing business practices as a form of consumer effort to minimize the impacts caused in terms of social, economic, and ecological aspects (Bundgaard & Huulgaard, 2019; Cooper, 2016; Ertz et al., 2019), as well as to ensure the recovery of products after their lifecycle and minimize waste (Darby & Obara, 2005). Businesses incorporating sustainable consumption into their strategy are now more popular among consumers (Bianchi & Birtwistle, 2012). Businesses can establish emotional bonds with customers by providing services that address their demands for dropping goods and participating in environmental improvement projects (Paden & Stell, 2005).

This study investigates the post-consumption area of apparel, focusing on how consumers dispose of clothes that certain parties no longer use. The research below offers a new and more thorough approach to this subject by forecasting consumer disposal behavior in both consumption dimensions and the relatively unexplored area of clothing disposal.

 Received: November 10, 2024 | Accepted: December 18, 2024

Reusing old clothing through appropriate channels, such as reselling, giving to friends or family, or donating it to a charity, are examples of alternative disposal strategies that can prolong the life of used clothing while enhancing resource utilization, cutting residue, and promoting a sustainable circular impact on the environment (Degenstein et al., 2020; Laitala & Klepp, 2015; McNeill et al., 2020). Customers can dispose of their clothing in three different ways, according to Bianchi & Birtwistle (2010): give it to their friends or family, donate it to charitable organizations (philanthropic behavior), and sell it (monetary behavior).

In addition, it is essential to remember that societal norms strongly influence proper disposal behavior. Sustainable practices such as recycling have barriers due to the collectivist culture found in many Asian countries (Nigbur et al., 2010) and influence households to behave in environmentally friendly ways by not littering (Zhang et al., 2023). In addition, there are subjective norms that play an important role. Both norms can support eco-friendly behavior such as garment disposal. The subjective norm from others and family members affects disposal behavior (H. Joung & Park-Poaps, 2011; Tang et al., 2010). In other studies, subjective norms do not affect behavioral intentions (Jena & Sarmah, 2015; Tonglet et al., 2004). Thus, social norms and peer behavior seem to have less of an impact on dumping behavior (H. M. Joung & Park-Poaps, 2013).

In addition to norms, environmental awareness is another important factor influencing consumer behavior. Environmentally friendly people are inclined to redistribute clothing responsibly. Research indicates that the following elements significantly influence disposal practices: sustainability, environmental consciousness, willingness to contribute, and individual characteristics (Harris et al., 2016; Paço et al., 2020). According to Yee et al. (2016), increasing environmental awareness and providing clear information about disposal routes and their environmental implications are crucial to encourage sustainable clothing disposal behavior. However, other research has shown that environmental awareness does not automatically translate into sustainable behavior; the consumer's attitude plays a vital role in determining disposal techniques (Nanayakkara, 2019; Yee et al., 2016).

There is still no inconsistency in previous studies and a limited understanding of how consumers choose among various disposal techniques, including philanthropic disposal behavior (donating, giving) and economical disposal behavior (reselling). This gap suggests a need for more research on strategic sustainable disposal practices and the factors influencing these decisions (Cerio & Debenedetti, 2021; De Ferran et al., 2020). This study intends to conduct additional research to validate existing research on how norms and environmental concerns can influence a person's post-consumption behavior, especially in more comprehensive sustainable disposal choices.

By investigating the motivations behind sustainable disposal practices and the impact of various factors, this study offers knowledge that can guide stakeholders toward more sustainable clothing disposal practices (L. Chen & Chang, 2020; Hassan et al., 2022; Vlastelica et al., 2023). Providing knowledge on the supporting elements of post-consumption practices through more responsible clothing disposal methods and new perspectives on making sustainable promotion efforts more effective are part of the main objectives of this research. This study intends to investigate the direct impact of social norms, subjective norm, and environmental awareness on philanthropic (charitable donations, gifts to family or friends) and economic (resell) clothing disposal behaviors.

2. Literature Review

2.1. The Theory of Interpersonal Behavior (TIB)

This represents a complex and multifaceted issue. A behavior reaction in any context results from multiple aspects, encompassing a person's objectives, habitual responses, contextual constraints, and other intrinsic elements of the circumstance (Egmond & Bruel, 2007) Their interpretation of appropriateness shapes individual behavior in a specific context. The theories of reasoned action (Fishbein & Ajzen, 1977) and planned behavior (Ajzen & Fishbein, 1980) are developed on the basis of the theory of interpersonal behavior (Triandis, 1977), which is adopted in this work. The TIB is especially adept at elucidating and analyzing intricate human behaviors within socially relevant circumstances. Social, affective, habitual, and conditional components are important aspects of changing a person's intentions and behavior (Triandis, 1977). This theory concludes that people's thoughts and feelings are the most critical aspects in determining their actions (Triandis, 1977; White et al., 2019). TIB is gaining prominence in the study of the factors influencing people’s ability to behave in an eco-friendly way (Issock Issock et al., 2020; Sung et al., 2019).

2.2. Social Norms

Social norms are generally agreed-upon expectations for how people in a particular group should act and respond to specific situations (Pristl et al., 2021). Numerous areas of sustainable behavior benefit from social norms (Doran & Larsen, 2016; Hanss et al., 2016; Pristl et al., 2021). Research shows that individuals are more inclined to do their part of the environment when they receive messages indicating that their nearby neighborhoods are equally committed to sustainability (Doran & Larsen, 2016). Borusiak (2020) clarified that people typically experience peer pressure to act in a certain way.

2.3. Subjective Norms

What constitute "subjective norms" are the forces that push people to act in a certain way (Ajzen, 1991). Subjective norms have been taken into account in recycling research to examine human behavior (Echegaray & Hansstein, 2017; Wang et al., 2016). Multiple studies have demonstrated how subjective norms serve as a vital measure of return intent (M. Chen & Tung, 2010; Wan et al., 2014). Subjective standards are employed to ascertain which activities (donating, reselling, and gifting to family or friends) are most influenced, enhancing human behavior comprehension.

2.4. Environmental Awareness

Being conscious of the environment plays a significant role in shaping people's behavior. Environmental awareness strongly impacts disposal behavior (Zalpuri et al., 2022). Supported by Goodman et al. (2020), reducing waste will not be practical without awareness of environmental impacts. This is consistent with the results of Ssemugabo (2020). This indicates that people's awareness of the dangers of improperly handling waste motivates them to follow proper trash disposal techniques. Littering is less common among those who are more aware of the damage that it does to the environment (Miller & Burbach, 2020). The incorporation of environmental awareness into daily practices can result in significant waste reduction and a smaller ecological footprint (Manucom et al., 2023).

2.5. Philanthropic Clothing Disposal Behavior (Donate, give to family or friend)

Philanthropic acts are voluntary and reflect a commitment to social responsibility (S. Kim & Jang, 2023). Philanthropic behavior is recognized as an important aspect of sustainable consumption practices (L. Chen & Chang, 2020). Altruism, awareness, environmental and social knowledge, economic factors, and psychological aspects influence philanthropic behavior (L. Chen & Chang, 2020; Hwang et al., 2020; Yee et al., 2016). Instead of just throwing away the clothing they purchase, consumers are more inclined to donate or give it away (Domina & Koch, 1999; Park et al., 2016). Another example of philanthropic clothing behavior is donating or trading with family and friends for valuable, unsuitable, or unused clothing (Bianchi & Birtwistle, 2012; H. M. Joung, 2014; Koch & Domina, 1999). However, not everyone will part with their expensive, unneeded clothing (Bianchi & Birtwistle, 2012). The guilt of throwing away old garments, plus factors including fit, worth, connection, and aesthetics, can motivate some people to save them (H. M. Joung, 2014).

2.6. Economic Clothing Disposal Behavior (Resell)

Selling old, unwanted clothes is an example of economical disposal behavior. This is because the person feels bad about throwing out their clothes. Consequently, they sell surplus clothing, which may also facilitate the generation of recurring revenue while safeguarding the environment (Hassan et al., 2022). Economic considerations have motivated individuals to resale and repurpose garments (H. M. Joung & Park-Poaps, 2013). Interestingly, individuals interested in fashion use alternative disposal methods such as reselling, bartering, or participating in return programs (Weber et al., 2017). In addition, although environmental attitudes influence clothing disposal behavior, they are not always manifested in sustainable practices unless accompanied by a positive attitude toward economical disposal methods (Yee et al., 2016).

2.7. The Effect of Social Norms on Philanthropic Clothing Disposal Behavior (Donate, Give to Families or Friends)

Individuals tend to be more involved in charitable giving when they feel that their peers are also donating. For example, Agerströmet et al. (2016) reported that local descriptive norms significantly increased charitable donations, suggesting that individuals were motivated to conform to the behavior of those around them. Everett (2014) proposed that the tendency to follow social norms can be used to encourage more altruistic behavior. People are more inclined to donate when they know that their colleagues have done so, as demonstrated by Nook (2016). This suggests that social norms not only guide individual behavior but also create a feedback loop where generous actions by one individual can inspire similar behavior in others. Joung (2011) Additionally, social norms affect college students' garment disposal behavior, suggesting family and friend attitudes can encourage more sustainable disposal methods.

Research shows that personal norms, shaped by individual values and beliefs, interact with social norms to influence behavior (Yamin et al., 2019). People give their things to charities, family, and friends, sell them to secondhand stores, or throw them away (Bianchi & Birtwistle, 2012; H. M. Joung, 2014; Koch & Domina, 1999). Consequently, we formulate the following hypothesis:

H1: Social norms positively impact philanthropic disposal behavior (donate to charity).
H2: Social norms positively impact philanthropic disposal behavior (giving away to families or friends).

2.8. The Effect of Social Norms on Economic Clothing Disposal Behavior (Resell)

Additionally, selling unwanted clothes is often linked to social and cultural factors. Studies show that young, urban consumers, particularly women, are more inclined to resell or donate clothing. This demographic trend underlines how disposal habits are shaped by social norms within particular populations (Mcneill et al., 2020; Sonnenberg et al., 2019). Liu (2024) claims that traditional beliefs and modern fashion trends impact consumers' desire to resell to fit social norms. Social media

dramatically impacts how people behave when they resell. People can discuss their resale activities and experiences and interact with like-minded customers via social media. Visibility fosters community and reinforces resale as good behavior (S. Liu et al., 2024). After seeing their friends resell products and receive favorable comments, they may be inspired to normalize the practice in their social circles. Laitala's synthesis of research results indicates that consumer decisions regarding clothing disposal are influenced by various factors, including environmental awareness, social norms, and economic considerations (Laitala, 2014).

H3: Social norms positively influence economic disposal behavior (resell)

2.9. The Effect of Subjective Norms on Philanthropic Disposal Behavior (Donate, Give to Families or Friends)

Individuals tend to emulate their peers and relatives. Environmental motivation can affect recycling behavior (Shim, 1995). Motivation guides behavior (Hawkins et al., 1988). Family and peers influence clothing-throwing behavior as well as social media. Research has shown that support from family and friends makes people more likely to wear clothes (Silva et al., 2021). This family influence can create a supportive environment where donating clothes is perceived as a shared value, reinforcing behavior. For example, if a family regularly donates clothes to charities or distributes them to relatives, younger members may adopt this practice as part of their socialization process (Melnyk et al., 2021). The following hypotheses were formulated to determine which behaviors are most significantly influenced by subjective norms, as previous research has not sufficiently investigated these behaviors:

H4: Subjective norms positively affect philanthropic disposal behavior (donate to charity)

H5: Subjective norms positively affect philanthropic disposal behavior (give away to family or friends)

2.10. The Effect of Subjective Norms on Economic Disposal Behavior (Resell)

Numerous studies have shown that subjective norms affect resale intentions. Subjective norms are positively correlated with clothing resale and donation intent (Goudeau & Lee, 2022). This finding demonstrates that people's propensity to resell increases when they believe that their peers do. Yee et al. (2016) reported that customers with good attitudes toward clothing disposal are likely to engage in sustainable behavior, supporting the premise that subjective standards can influence behavior. Vlastelica et al. (2023) demonstrated that subjective standards may not always be the most important predictors of discarding intentions, but they impact them, especially in circumstances with high social pressure. The theory of planned behavior acknowledges subjective norms as drivers that influence behavioral changes in individuals (Ko & Jin, 2017). This shows that consumers are driven by environmental concerns and the economic benefits of reselling items. Clothing as a valuable asset that may be resold or donated correlates with the expanding trend of secondhand purchasing, which has gained popularity among consumers seeking economic and environmental benefits (Sumo, 2022). Fashion interest also shows how subjective norms affect the disposal of economic clothes. Conversely, shoppers who tend to be more drawn to fashion tend to resell the clothes they have rather than throw them in the (Weber et al., 2017). Similarly, Koay et al. (2024) discovered how subjective standards, behavioral beliefs, and perceived control influence customers' secondhand clothing purchases.

H6: Subjective norms positively affect economic clothing disposal behavior (resell)

2.11. The Effect of Environmental Awareness on Philanthropic Disposal Behavior (Donate to Charity, Give to Families or Friends)

The findings suggest that the motivation to engage in charitable donations is closely tied to an individual's environmental consciousness, which is further reinforced by their general recycling behaviors (Aydın, 2017). The complexity of clothing disposal behavior is further underscored by Ekström & Salomonson (2014), who argue that understanding the link between clothing consumption and environmental impact is essential for promoting reuse and recycling. This perspective aligns with the notion that increased environmental awareness can lead to more responsible disposal behaviors, including charitable donations. According to Lundberg et al. (2019), environmental awareness increases philanthropic contributions. Donating to secondhand stores may indicate an increasing tendency toward sustainability among younger buyers (Norum, 2015). Hassan et al. (2022) reported that sustainable fashion consumption increases environmental awareness and charitable disposal (donating and giving).

H7: Environmental awareness positively affects philanthropic disposal behavior (donate to charity)

H8: Environmental awareness positively affects philanthropic disposal behavior (give to family or friends)

2.12. The Effect of Environmental Awareness on Economic Disposal Behavior (Resell)

Studies show that environmentalists are more likely to act sustainably. For example, reselling clothes can help reduce waste and promote a circular economy (C. Liu et al., 2023). When consumers have high environmental awareness, they are more likely to recycle without needing additional monetary incentives. This behavior supports optimal production plans for firms, reducing the need for financial incentives to encourage recycling (Jiang & Zheng, 2023). This implies that environmentally aware consumers might prefer reselling items as part of their recycling efforts. People who label themselves as environmentalists are more likely to resell their items rather than throw them away because of their positive attitudes toward

recycling practices (Handriana & Ningsih, 2021). This shift in mindset reflects a growing trend toward sustainability, where consumers are concerned about the initial purchase, the product's lifecycle, and its resale potential.

H9: Environmental awareness positively affects economic disposal behavior (resell)

Figure 1 outlines the research paradigm that we used. Referring to Figure 1, we determined the causes and effects of environmental awareness, subjective norms, and societal norms regarding the resale of used apparel for charitable purposes.

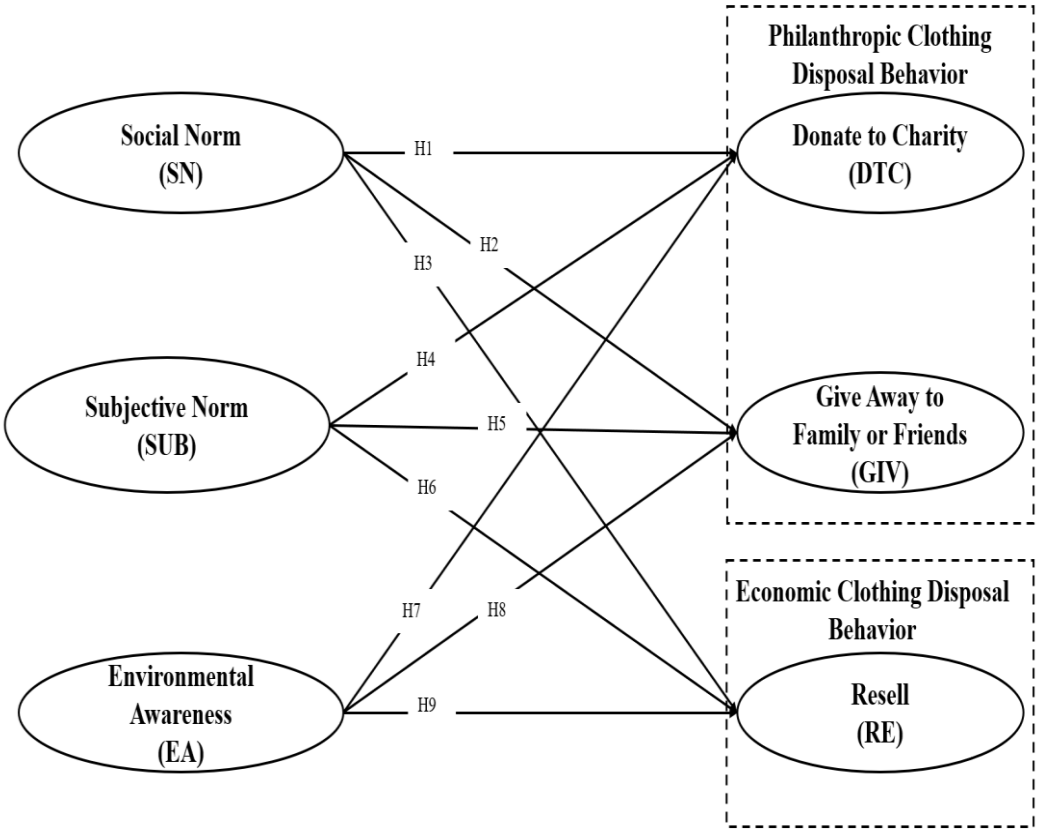


Figure 1 Research Framework.

3. Methods

Quantitative methodologies and explanatory survey design. The responders are apparel-focused fashionistas. This study targeted Indonesian 18-35 year olds. This study needs a minimum sample size of 130-260 people. This is because we follow the sample determination rules determined by Hair (2011). The total number of statement items must be multiplied by at least five and a maximum of ten. Primary data are collected via Google form URLs. A total of 251 of 279 questionnaire responses were suitable for analysis. Bootstrap analysis was used to test our model on AMOS 29.

The product category for this study is clothing. This study's exogenous variables include social norms, subjective norms, and environmental awareness. The four social norm indicators were from Hassan et al. (2022), whereas the five subjective norm indicators came from Khan et al. (2019) and Bianchi and Birtwistle (2010). Four Moon & Lai (2015) indicators measure environmental awareness. Two things about people who donate their old clothes are endogenous factors in this study. One dimension is given to charity (four indicators from Wai Yee et al. (2016) And Khan et al. (2019), and the other is given to family or friends (four indicators from Hassan et al. (2022). Four variables were employed by Khan et al. (2019) To quantify the second endogenous variable, economic disposal behavior (resell). From "strongly disagree" (point 1) to "strongly agree" (point 7), the Likert scale offers seven possible responses.

4. Results

4.1. Preliminary Data Analysis

The respondents in this study were primarily female (68%), and most of our respondents were 18-22 years old (59%) or 23-27 years old (33%). Therefore, the criteria for young adult respondents have been met. The leading educational qualifications are a bachelor's degree (44%) and senior high school (41%), with an average income below 2.500.000,00 Rupiah (74%). The respondents in this study were primarily female (68%), and most of the respondents were aged 18-22 years (59%) or 23-27 years (33%), as shown in Table 1.

Table 1 Sample characteristics.

Respondents Characteristic	Information	Frequency	Percentage (%)
Gender	Male	80	32
	Female	171	68
Age	18-22	149	59
	23-27	82	33
	28-32	18	7
	33-35	2	1
Education	Junior	0	0
	Senior High School	103	41
	Diploma	11	4
	Bachelor	111	44
	Postgraduate	26	10
Income	2.500.000	186	74
	2.500.000-5.000.000	57	23
	5.000.000-7.500.000	8	3
	7.500.000-10.000.000	0	0

Table 2 shows our instrument validity and reliability tests. Table 2 shows that the loading factors are still less than 0.5; therefore, we removed the invalid items. The remaining items had loading factors greater than 0.5, indicating validity. When the results of the study show factor loadings of more than 0.5, reliability of more than 0.7, and AVE values of more than 0.5, the data are valid and meet the criteria for further testing (Hair et al., 2011). The analysis revealed that the research tool has validity and reliability.

Table 2 Validity and Reliability.

Variable	Indicators	Factor Loading	CR	AVE	DV
Social Norms (X1)	Family members I respect will approve of my pro-environmental behavior.	0.824	0.925	0.755	0.869
	Family members I respect will approve of my eco-friendly clothing purchases.	0.888			
	A dear friend will fund my eco-friendly clothing purchases.	0.922			
	The public will support my eco-friendly apparel purchases.	0.839			
Subjective Norm (X2)	Most of my loved ones support recycling plastic debris.	0.781	0.906	0.707	0.841
	Most of my loved ones would approve of my trash recycling.	0.853			
	Most of my friends support recycling trash.	0.817			
	Most of my loved ones want me to recycle.	0.869			
	My family supports waste recycling.	0.824			
Environmental Awareness (X3)	I worry about the environment.	Deleted	0.826	0.624	0.790
	Destroying the environment makes me angry.	0.535			
	I advise my friend to use eco-friendly products.	0.897			
	I buy products in environmentally safe packaging.	0.883			
Donate To Charity (Y1)	I give my stuff to charity to save the earth.	0.801	0.866	0.617	0.785
	I give my items to charity.	0.782			
	Giving my belongings to charity makes me joyful.	0.789			
	I try to give what I have to help.	0.769			
Give Away to Family or Friends (Y2)	I transferred old stuff to my family.	Deleted	0.888	0.798	0.893
	I gave a friend my old stuff.	Deleted			
	I exchanged clothes with family members.	0.873			
	I exchanged clothes with a friend.	0.913			
Resell (Y3)	I sell unwanted stuff for reduced trash.	0.840	0.899	0.691	0.831
	I sell secondhand stuff for profit.	0.740			
	I am selling my good-condition secondhand stuff.	0.930			
	I save money by exchanging used items at thrift stores.	0.804			

Figure 2 presents the framework that was processed via SEM AMOS 29, and as shown, we combine three exogenous variables: social norms, subjective norms, and environmental awareness. The three endogenous variables include philanthropic clothing disposal behavior, which includes acts such as gifting to friends and relatives or charities, and the economical disposal of garments, represented by resale operations.

The next step is to ensure that the model we test meets the goodness-of-fit principle with the specific criteria presented in Table 3. The loading factor values for the research variable indicators are greater than 0.5, except for EA1, GIV1, and GIV2, and need to be deleted, as shown in Figure 2.

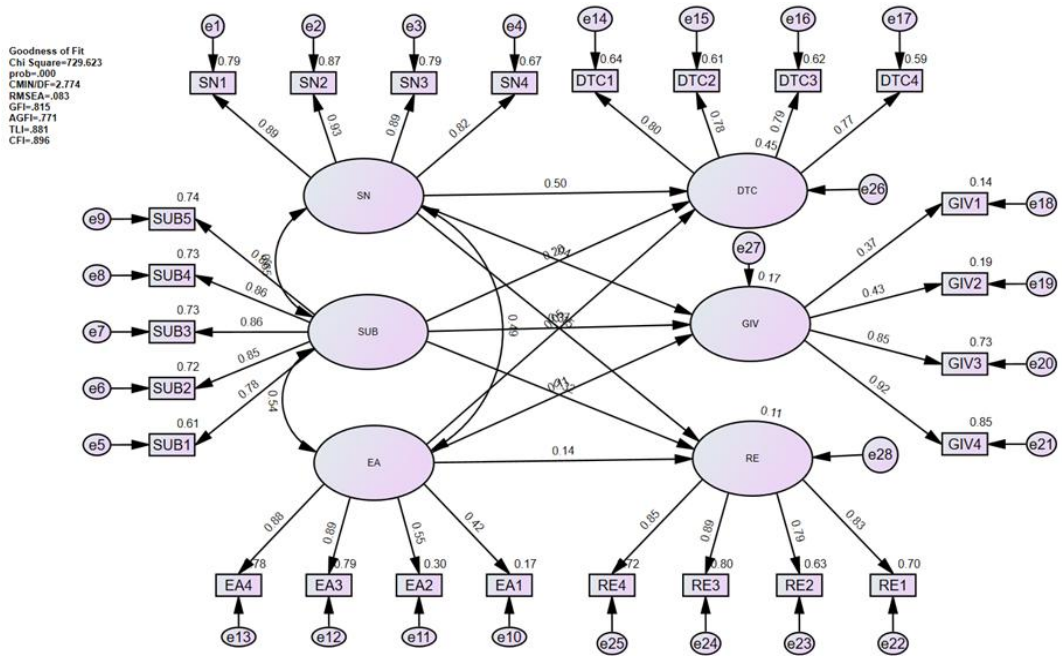


Figure 2 First Model Test.

Table 3 First Model Goodness of Fit (GOF).

The Goodness-of-Fit Index	Cutoff Value	Result	Model Evaluation
Chi-Square	Expected Small	729.623	Not Fit
Significant Probability	≥ 0.05	0.000	Not Fit
GFI	≥0.90	0.815	Marginal
RMSEA	≤0.08	0.083	Not Fit
CMIN/DF	≤2.00	2.774	Not Fit
AGFI	≥0.90	0.771	Not Fit
TLI	≥0.90	0.881	Marginal
CFI	≥0.90	0.896	Marginal

The GOF Full Model value presented in Table 3 is a criterion for assessing model fit. The model remains unfit, as none of the goodness-of-fit statistic (GOF) values meet the assessment criteria. MI improves the goodness of fit (GOF) to satisfy standards. The covariance values of the mistakes from each research indicator with the highest MI value form a causal association line in the MI.

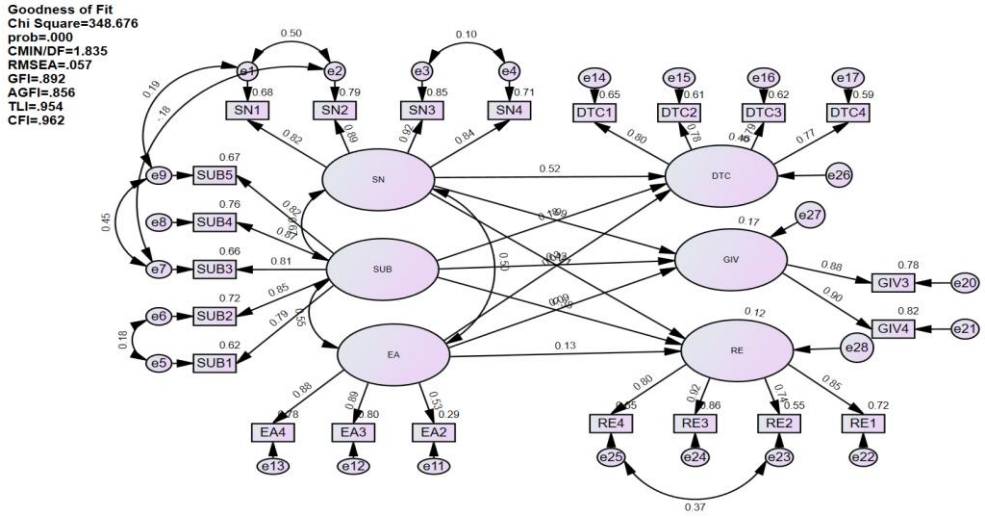


Figure 3 Final Model Test.

Following the elimination of insignificant items, it is evident that all the research variable indicators present loading factor values exceeding 0.5, as illustrated in Figure 3. The final test results show that the five GOF criteria have been met and

are ready for further testing, as shown in Table 4 below. Consequently, we determined that our model could serve as a framework for hypothesis testing analysis.

Table 4 Final Model Test Goodness of Fit (GOF).

The Goodness-of-Fit Index	Cutoff Value	Result	Model Evaluation
Chi-Square	Expected Small	348.676	Fit
Significant Probability	≥ 0.05	0.000	Not Fit
GFI	≥0.90	0.892	Marginal
RMSEA	≤0.08	0.057	Fit
CMIN/DF	≤2.00	1.835	Fit
AGFI	≥0.90	0.856	Marginal
TLI	≥0.90	0.954	Fit
CFI	≥0.90	0.962	Fit

Multivariate normal data are estimated via maximum likelihood (ML), and data that do not follow a normal distribution are estimated via bootstrap maximum likelihood after the GOF test. Table 5 shows that the research data still do not meet normal standards. Multivariate normally distributed data have skewness or kurtosis values between -2.58 and 2.58. The data are not normally distributed, so a bootstrap resampling procedure with 500 resamplings and a bias-corrected confidence interval of 95% was used on the basis of AMOS normality results.

The variance inflation factor (VIF) is a standard diagnostic tool for detecting multicollinearity, with a threshold typically set at 5-10 (Greene, 2003; J. H. Kim, 2019). Other approaches also imply that the VIF of each construct should be lower than 5 (Hair et al., 2011). Multicollinearity is typically detected via regression models where independent variables are strongly correlated (Pereira et al., 2020). SPSS 23 tools were used to test for multicollinearity. The result is that all the independent variables' VIF values are less than 10. We conclude that the framework used in this analysis meets the multicollinearity assumption. It can be seen in Table 6.

Table 5 Normality Test.

Variable	min	max	skew	c.r.	kurtosis	c.r.
RE4	1.000	7.000	-.408	-2.673	-.781	-2.559
RE3	1.000	7.000	-.127	-.830	-1.086	-3.561
RE2	1.000	7.000	-.484	-3.175	-.793	-2.601
RE1	1.000	7.000	.054	.356	-1.028	-3.370
GIV4	1.000	7.000	-.336	-2.206	-.910	-2.984
GIV3	1.000	7.000	-.528	-3.464	-.722	-2.367
DTC4	3.000	7.000	-1.176	-7.712	.915	3.001
DTC3	3.000	7.000	-.972	-6.371	.070	.230
DTC2	4.000	7.000	-1.300	-8.525	1.090	3.574
DTC1	3.000	7.000	-1.270	-8.329	1.076	3.529
EA4	2.000	7.000	-.539	-3.534	-.444	-1.457
EA3	3.000	7.000	-.536	-3.514	-.696	-2.282
EA2	2.000	7.000	-.723	-4.742	.017	.054
SUB5	2.000	7.000	-.997	-6.541	.089	.292
SUB4	2.000	7.000	-.439	-2.878	-.966	-3.168
SUB3	3.000	7.000	-.884	-5.794	-.143	-.469
SUB2	3.000	7.000	-.598	-3.925	-.744	-2.441
SUB1	2.000	7.000	-.536	-3.517	-.627	-2.055
SN4	3.000	7.000	-.960	-6.293	-.152	-.499
SN3	3.000	7.000	-1.053	-6.907	.254	.834
SN2	3.000	7.000	-1.018	-6.678	.037	.121
SN1	4.000	7.000	-.994	-6.516	.087	.284
Multivariate					115.379	28.515

Table 6 Collinearity.

Independent Variable	Variance Inflation Factor (VIF)
Social Norms	1,714
Subjective Norms	1,735
Environmental Awareness	1,359

Table 7 shows the hypothesis testing outcomes in the next section. The study's structural model's variable relationships are tested via hypothesis testing. The bootstrap two-tailed significance section presents the hypothesis test findings and variable influence coefficients. The chance of accepting the hypothesis is calculated using p values smaller than the alpha value of 0.05.

Table 7 Hypothesis Testing.

			P	Hypothesis
Social Norms	→	Donate to Charity (DTC)	0.004	Supported
Social Norms	→	Give Away to Family or Friends (GIV)	0.347	Not Supported
Social Norms	→	Resell (RE)	0.015	Supported
Subjective Norms	→	Donate to Charity (DTC)	0.031	Supported
Subjective Norms	→	Give Away to Family or Friends (GIV)	0.001	Supported
Subjective Norms	→	Resell (RE)	0.001	Supported
Environmental Awareness	→	Donate to Charity (DTC)	0.687	Not Supported
Environmental Awareness	→	Give Away to Family or Friends (GIV)	0.436	Not Supported
Environmental Awareness	→	Resell (RE)	0.188	Not Supported

Table 6 summarizes the hypothesis testing outcomes. Hypothesis 1 examines the associations between the variables, with a probability of 0.004, which is less than 0.05. Thus, "Social norms positively affect philanthropic clothing disposal behavior (donate to charity)" supports Hypothesis 1. Second, Hypothesis 2 demonstrated that 0.347 is more significant than 0.05 for the estimated parameter probability. This does not support Hypothesis 2, which claims that social norms encourage philanthropic disposal (giving to family or friends). Third, our hypothesis 3 test yielded a probability value of 0.015 for the link between the two variables, which is less than 0.05. Thus, Hypothesis 3, that societal norms positively impact economic disposal (resell), is supported. Fourth, Hypothesis 4 demonstrated that 0.031 was less than 0.05 for the associations between the variables. We support Hypothesis 4, which argues that subjective norms positively promote philanthropic disposal (donate to charity). Fifth, with respect to Hypothesis 5, an examination of the association between two variables reveals that 0.001 is less than 0.05. Thus, Hypothesis 5 states that subjective norms positively affect charitable clothing disposal (giving to family or friends). Sixth, Hypothesis 6 demonstrated that 0.001 was less than 0.05 for the associations between the variables. Thus, Hypothesis 6, that subjective norms positively impact economic clothing disposal (resell), is supported. Recently, Hypothesis 7 showed that 0.687 was more than 0.05 for the association between two variables. Our hypothesis that environmental awareness increases philanthropic garment disposal (donate to charity) is not supported. Eighty, Hypothesis 8, tested the associations between the variables and found that 0.436 was greater than 0.05. Thus, Hypothesis 8, which states that environmental awareness increases altruistic clothing disposal (giving to family or friends), is unsupported. Hypothesis 9 has a p value of 0.188, which exceeds the threshold of 0.05. The data presented do not support Hypothesis 9, which posits that environmental awareness influences the disposal of commercial garments through resale.

5. Discussion

The findings of the first and third hypothesis tests indicate that social norms positively and significantly influence philanthropic disposal behavior, especially donating to charity and economical disposal behavior (resell). These findings prove that social expectations significantly impact individuals' charitable actions. The possibility that someone will donate increases with social expectations. This aligns with the findings of Knowles et al. (2012) and Layton (2014). In contrast, the influence of social norms on giveaways to family or friends appears to be weaker. Hassan et al. (2022) noted that while social norms can influence donation frequency, sustainable consumption often plays a mediating role. Furthermore, subjective norms positively affect philanthropic and economic disposal. Consumers with environmentally concerned family, friends, and colleagues may experience pressure to adopt pro-environmental behaviors because the information indicates that those in their social circle are engaging in similar actions (Borusiak et al., 2020; Doran & Larsen, 2016). This is particularly applicable to young adult customers, who highly prioritize the opinions of their peers. Our seventh to ninth hypothesis test results show that environmental awareness influences philanthropic and economical disposal behavior insignificantly. While it is commonly assumed that increased ecological awareness leads to more pro-environmental behaviors, the effectiveness of this awareness is often moderated by cultural and social factors, as seen in Mehta's (2023) results emphasize that buying sustainably does not always follow from environmental knowledge. Social norms do not affect family or friend giveaways. None of the dependent variables are modified by environmental awareness. This study shows that humanitarian and economical disposal activities should raise subjective standards rather than social or ecological awareness.

According to this study, young adult consumers respect peer comments. Establishing a referral program and using client testimonials to make it seem natural can reward consumers who refer friends and family. Previous researchers also agree that peer influence plays a crucial role in shaping sustainable purchasing behaviors among young adults (Blose et al., 2020; Elgammal et al., 2024). According to subjective and social norms, clients choose more sustainable disposal methods, including donating to friends and family or reselling. Businesses can increase earnings by offering discounts to clients who donate old clothes for each new item and permanent membership points or special privileges to repeat donors. This is supported by previous findings, which revealed that subjective norms significantly impact attitudes toward sustainable consumption and donation behavior. This influence is particularly noted among young adults, where subjective norms directly encourage donation disposal behavior (Maulani et al., 2024). Additionally, subjective norms were identified as a key determinant of sustainable fashion

consumption intentions and behaviors (Yang et al., 2024). Similarly, social norms were highlighted as a significant factor in shaping young consumers' sustainable fashion consumption and clothing disposal patterns (Hassan et al., 2022) and influencing their intentions and actual behaviors toward sustainable fashion (Hong et al., 2024). Since norm have a significant impact, brands can provide consumers with clothing care and disposal recommendations on packaging or websites to help normalize sustainable practices, making them a standard part of consumer behavior. Then, clear instructions on how to dispose of clothing responsibly, can help reduce textile waste (Ahmed, 2020; McNeill et al., 2020).

These findings suggest philanthropic organizations and NGOs grow their programs by developing local volunteer groups and school or university-based campaigns to build subjective norms among young people whom peers easily influence. Positive peer norms are associated with increased prosocial behaviors and reduced antisocial behaviors (McKeown & Taylor, 2018). Other findings said that Sustainable village projects can significantly benefit from community involvement and education (Udin et al., 2024). In this case, governmentally sustainable village projects may help communities donate and recycle old clothes by educating residents on the importance of sustainable disposal and providing practical workshops.

Despite its importance, this study contains several issues that must be resolved in follow-up investigations. Their narrow geographical reach prevented the broad application of these findings. Indonesian youth aged 18–35 years are the focus of this study on social norms, subjective norms, and environmental awareness in sustainable disposal behaviors. Thus, future research must integrate socioeconomic status, education, and gender roles in the conclusions of this study. Future sustainability research could apply social practice ideas to the sharing economy and smart cities. Afterwards, this study concentrated on quantitative methodologies. Interdisciplinary research using qualitative methods to understand customer preferences and behaviors is needed.

6. Conclusion

This study used the Theory of Interpersonal Behavior (TIB) to investigate the factors influencing sustainable disposal behavior among young adult consumers in Indonesia, including philanthropic (donate, give to family or friends) and economic (resell) disposal of apparel. Overall, the findings show that norms have a direct and significant impact on sustainable disposal practices. However, it differs slightly from social norms, which do not significantly contribute to family giving behavior, and environmental awareness, which has no significant influence on respondents' decision to engage in sustainable clothes disposal behavior. This demonstrates that, despite their high environmental awareness, young adult consumers in Indonesia have yet to fully mobilize themselves to adopt a more sustainable method of clothing disposal. Unlike when they are affected by their surroundings or peers, individuals are more easily inspired to act sustainably when they are supported and see other doing the same.

Ethical considerations

Several ethical considerations were considered in this study to maintain the integrity of the research process and protect participants' rights. The study begins with the explicit consent of each participant before completing the online questionnaire by first asking about their willingness. The participants were informed about the purpose, questions, duration, and freedom to withdraw without penalty. These facts will be explained in writing to help participants understand their position in the study. The anonymity and confidentiality of the participants will be protected. No personal information about the participants was collected. The data collected will be anonymized and stored securely. These responses will only be used for this study and will not be shared.

Conflict of interest

The authors declare no conflicts of interest.

Funding

This research did not receive any financial support.

References

Agerström, J., Carlsson, R., Nicklasson, L., & Guntell, L. (2016). Using descriptive social norms to increase charitable giving: The power of local norms. *Journal of Economic Psychology, 52,* 147–153. https://doi.org/10.1016/j.joep.2015.12.007

Ahmed, M. U. (2020). Consumer engagement and roles for sustainable textiles. In *Sustainable Textile and Fashion Value Chains: Drivers, Concepts, Theories and Solutions* (pp. 237–253). https://doi.org/10.1007/978-3-030-22018-1_13

Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes, 50*(2), 179–211.

Ajzen, I., & Fishbein, M. (1980). *Understanding attitudes and predicting social behavior.* Prentice-Hall Inc.

Aydın, H. (2017). Used clothing disposal behavior within the scope of sustainable consumption. *International Journal of Academic Research in Economics and Management Sciences.* https://doi.org/10.6007/ijarems/v6-i3/3148

Bianchi, C., & Birtwistle, G. (2010). Sell, give away, or donate: An exploratory study of fashion clothing disposal behaviour in two countries. *International Review of Retail, Distribution and Consumer Research, 20*(3), 353–368. https://doi.org/10.1080/09593969.2010.491213

Bianchi, C., & Birtwistle, G. (2012). Consumer clothing disposal behaviour: A comparative study. *International Journal of Consumer Studies, 36*(3), 335–341. https://doi.org/10.1111/j.1470-6431.2011.01011.x

Blose, J. E., Mack, R. W., Pitts, R. E., & Xie, H. Y. (2020). Exploring young U.S. and Chinese consumers' motivations to recycle. *Journal of International Consumer Marketing, 32*(1), 33–46. https://doi.org/10.1080/08961530.2019.1618776

Borusiak, B., Szymkowiak, A., Horska, E., Raszka, N., & Zelichowska, E. (2020). Towards building sustainable consumption: A study of second-hand buying intentions. *Sustainability (Switzerland), 12*(3), 1–15. https://doi.org/10.3390/su12030875

Brewer, M. K. (2019). Slow fashion in a fast fashion world: Promoting sustainability and responsibility. *Laws.* https://doi.org/10.3390/laws8040024

Bundgaard, A. M., & Huulgaard, R. D. (2019). Luxury products for the circular economy? A case study of Bang & Olufsen. *Business Strategy and the Environment, 28*(5), 699–709.

Cerio, E., & Debenedetti, A. (2021). "Should I give it away or sell it?" A strategic perspective on consumers' redistribution of their unused objects. *Journal of Business Research, 135,* 581–591.

Chen, L., & Chang, C.-E. (2020). Clothing disposal behavior of Taiwanese consumers with respect to environmental protection and sustainability. https://doi.org/10.20944/preprints202010.0240.v1

Chen, M., & Tung, P. (2010). The moderating effect of perceived lack of facilities on consumers' recycling intentions. *Environment and Behavior, 42,* 807–823. https://doi.org/10.1177/0013916509352833

Cooper, T. (2016). The significance of product longevity. In *Longer Lasting Products* (pp. 3–36). Routledge.

Cudlínová, E., Buchtele, R., & Dušek, R. (2022). The fashion industry and its problematic consequences in the green marketing era: A review. *SHS Web of Conferences.* https://doi.org/10.1051/shsconf/202213501011

Darby, L., & Obara, L. (2005). Household recycling behaviour and attitudes towards the disposal of small electrical and electronic equipment. *Resources, Conservation and Recycling, 44*(1), 17–35.

De Ferran, F., Robinot, E., & Ertz, M. (2020). What makes people more willing to dispose of their goods rather than throwing them away? *Resources, Conservation and Recycling, 156,* 104682. https://doi.org/10.1016/j.resconrec.2020.104682

Degenstein, L. M., McQueen, R. H., McNeill, L. S., Hamlin, R. P., Wakes, S. J., & Dunn, L. A. (2020). Impact of physical condition on disposal and end-of-life extension of clothing. *International Journal of Consumer Studies, 44*(6), 586–596. https://doi.org/10.1111/ijcs.12590

Domina, T., & Koch, K. (1999). Consumer reuse and recycling of post-consumer textile waste. *Journal of Fashion Marketing and Management, 3*(4), 346–359. https://doi.org/10.1108/eb022571

Doran, R., & Larsen, S. (2016). The relative importance of social and personal norms in explaining intentions to choose eco-friendly travel options. *International Journal of Tourism Research, 18*(2), 159–166. https://doi.org/10.1002/jtr.2042

Echegaray, F., & Hansstein, F. V. (2017). Assessing the intention-behavior gap in electronic waste recycling: The case of Brazil. *Journal of Cleaner Production, 142,* 180–190.

Egmond, C., & Bruel, R. (2007). Nothing is as practical as a good theory. *Analysis of Theories and a Tool for Developing Interventions to Influence Energy-Related Behaviour.*

Ekström, K. M., & Salomonson, N. (2014). Reuse and recycling of clothing and textiles—A network approach. *Journal of Macromarketing, 34*(3), 383–399. https://doi.org/10.1177/0276146714529658

Elgammal, I., Ghanem, M., & Al-Modaf, O. (2024). Sustainable purchasing behaviors in Generation Z: The role of social identity and behavioral intentions in the Saudi context. *Sustainability (Switzerland), 16*(11). https://doi.org/10.3390/su16114478

Ertz, M., Leblanc-Proulx, S., Sarigöllü, E., & Morin, V. (2019). Made to break? A taxonomy of business models on product lifetime extension. *Journal of Cleaner Production, 234,* 867–880.

Everett, J. A. C., Caviola, L., Kahane, G., Savulescu, J., & Faber, N. S. (2014). Doing good by doing nothing? The role of social norms in explaining default effects in altruistic contexts. *European Journal of Social Psychology, 45*(2), 230–241. https://doi.org/10.1002/ejsp.2080

Fishbein, M., & Ajzen, I. (1977). *Belief, attitude, intention, and behavior: An introduction to theory and research.*

Goodman-Smith, F., Mirosa, R., & Mirosa, M. (2020). Understanding the effect of dining and motivational factors on out-of-home consumer food waste. *Sustainability, 12*(16). https://doi.org/10.3390/su12166507

Goudeau, C., & Lee, H.-J. (2022). Exploring the antecedents of apparel disposal behaviors. *International Journal of Research in Business and Social Science (2147-4478)*. https://doi.org/10.20525/ijrbs.v10i8.1517

Greene, W. (2003). *Econometric analysis* (5ª ed.).

Hair, J. F., Ringle, C. M., & Sarstedt, M. (2011). PLS-SEM: Indeed a silver bullet. *Journal of Marketing Theory and Practice, 19*(2), 139–151. https://doi.org/10.2753/MTP1069-6679190202

Handriana, T., & Ningsih, A. A. (2021). Green behavior and intention to charity on disposal of baby equipment products. *Review of International Geographical Education Online, 11*(4), 469–476.

Hanss, D., Böhm, G., Doran, R., & Homburg, A. (2016). Sustainable consumption of groceries: The importance of believing that one can contribute to sustainable development. *Sustainable Development, 24*(6), 357–370. https://doi.org/10.1002/sd.1615

Harris, F., Roby, H., & Dibb, S. (2016). Sustainable clothing: Challenges, barriers, and interventions for encouraging more sustainable consumer behaviour. *International Journal of Consumer Studies, 40*(3), 309–318. https://doi.org/10.1111/ijcs.12257

Hassan, S. H., Yeap, J. A. L., & Al-Kumaim, N. H. (2022). Sustainable fashion consumption: Advocating philanthropic and economic motives in clothing disposal behaviour. *Sustainability, 14*(3), 1875. https://doi.org/10.3390/su14031875

Hawkins, D. I., Best, R. J., & Coney, K. A. (1988). *Consumer behavior*. Irwin.

Hong, Y., Al Mamun, A., Yang, Q., & Masukujjaman, M. (2024). Predicting sustainable fashion consumption intentions and practices. *Scientific Reports, 14*(1), 1706. https://doi.org/10.1038/s41598-024-52215-z

Hwang, Y. Y., Jo, G. Y., & Oh, M. J. (2020). The persuasive effect of competence and warmth on clothing sustainable consumption: The moderating role of consumer knowledge and social embeddedness. *Sustainability, 12*(7). https://doi.org/10.3390/su12072661

Issock Issock, P. B., Roberts-Lombard, M., & Mpinganjira, M. (2020). Understanding household waste separation in South Africa: An empirical study based on

an extended theory of interpersonal behaviour. *Management of Environmental Quality: An International Journal, 31*(3), 530–547. https://doi.org/10.1108/MEQ-08-2019-0181

Jena, S. K., & Sarmah, S. P. (2015). Measurement of consumers' return intention index towards returning the used products. *Journal of Cleaner Production, 108*, 818–829.

Jiang, X., & Zheng, Y. (2023). Pricing decisions and remanufacturing strategies considering consumer recycling behavior. *Annals of Operations Research, 322*(2), 755–792.

Joung, H. M. (2014). Fast-fashion consumers' post-purchase behaviours. *International Journal of Retail and Distribution Management, 42*(8), 688–697. https://doi.org/10.1108/IJRDM-03-2013-0055

Joung, H. M., & Park-Poaps, H. (2013). Factors motivating and influencing clothing disposal behaviours. *International Journal of Consumer Studies, 37*(1), 105–111. https://doi.org/10.1111/j.1470-6431.2011.01048.x

Joung, H., & Park-Poaps, H. (2011). Factors motivating and influencing clothing disposal behaviours. *International Journal of Consumer Studies*. https://doi.org/10.1111/j.1470-6431.2011.01048.x

Khan, F., Ahmed, W., Najmi, A., & Younus, M. (2019). Managing plastic waste disposal by assessing consumers' recycling behavior: The case of a densely populated developing country. *Environmental Science and Pollution Research, 26*(32), 33054–33066. https://doi.org/10.1007/s11356-019-06411-4

Kim, J. H. (2019). Multicollinearity and misleading statistical results. *Korean Journal of Anesthesiology, 72*(6), 558–569. https://synapse.koreamed.org/articles/1156320

Kim, S., & Jang, Y. (2023). The estimates of philanthropic behavior and attitude in South Korea: Methodological consideration. *Journal of Philanthropy and Marketing, 28*(2), e1780.

Knowles, S. R., Hyde, M. K., & White, K. M. (2012). Predictors of young people's charitable intentions to donate money: An extended theory of planned behavior perspective. *Journal of Applied Social Psychology*. https://doi.org/10.1111/j.1559-1816.2012.00932.x

Ko, S. B., & Jin, B. (2017). Predictors of purchase intention toward green apparel products. *Journal of Fashion Marketing and Management, 21*(1), 70–87. https://doi.org/10.1108/jfmm-07-2014-0057

Koay, K. Y., Lim, W. M., Khoo, K. L., Xavier, J. A., & Poon, W. C. (2024). Consumers' motivation to purchase second-hand clothing: A multimethod investigation anchored on belief elicitation and theory of planned behavior. *Journal of Product & Brand Management, 33*(5), 502–515. https://doi.org/10.1108/jpbm-05-2023-4512

Koch, K., & Domina, T. (1999). Consumer textile recycling as a means of solid waste reduction. *Family and Consumer Sciences Research Journal, 28*(1), 3–17. https://doi.org/10.1177/1077727X99281001

Laitala, K. (2014). Consumers' clothing disposal behaviour: A synthesis of research results. *International Journal of Consumer Studies, 38*(5), 444–457. https://doi.org/10.1111/ijcs.12088

Laitala, K., & Klepp, I. G. (2015). Clothing disposal habits and consequences for life cycle assessment (LCA). In *Handbook of Life Cycle Assessment (LCA) of Textiles and Clothing*. Elsevier Ltd. https://doi.org/10.1016/B978-0-08-100169-1.00016-2

Lamb, D. T., Venkatraman, K., Bolan, N., Ashwath, N., Choppala, G., & Naidu, R. (2014). Phytocapping: An alternative technology for the sustainable management of landfill sites. *Critical Reviews in Environmental Science and Technology, 44*(6), 561–637. https://doi.org/10.1080/10643389.2012.728823

Layton, M. D., & Moreno, A. (2014). Philanthropy and social capital in Mexico. *International Journal of Nonprofit and Voluntary Sector Marketing, 19*(3), 209–219. https://doi.org/10.1002/nvsm.1498

Liu, C., Bernardoni, J. M., & Wang, Z. (2023). Examining Generation Z consumer online fashion resale participation and continuance intention through the lens of consumer perceived value. *Sustainability, 15*(10), 8213. https://doi.org/10.3390/su15108213

Liu, S., Liu, C., & Lang, C. (2024). Consumer engagement in fashion circularity in China: Exploring consumer online fashion resale through the lens of social practice theory. *Sustainability, 16*(11), 4733.

Lundberg, P., Vainio, A., Ojala, A., & Arponen, A. (2019). Materialism, awareness of environmental consequences and environmental philanthropic behaviour among potential donors. *Environmental Values, 28*(6), 741–762. https://doi.org/10.3197/096327119X15579936382527

Manucom, M. G. C., Alcaraz, K. P., Alejo, R. P. S., Gaddi, K. C. M., Recio, K. E. P., & Yamaguchi, R. G. (2023). Awareness of Generation Z students about The Plaf (Plastic Flamingo) and other campaigns concerning plastics in online shopping. *International Journal of Environment, Engineering and Education, 5*(1), 9–18.

Maulani, M. R., Fatmawati, I., & Nuryakin. (2024). Managing donation disposal behavior by assessing consumers' attitudes toward sustainable consumption. *E3S Web of Conferences, 571*. https://doi.org/10.1051/e3sconf/202457102008

McKeown, S., & Taylor, L. K. (2018). Perceived peer and school norm effects on youth antisocial and prosocial behaviours through intergroup contact in Northern Ireland. *British Journal of Social Psychology, 57*(3), 652–665. https://doi.org/10.1111/bjso.12257

McNeill, L. S., Hamlin, R. P., McQueen, R. H., Degenstein, L., Garrett, T. C., Dunn, L., & Wakes, S. (2020). Fashion sensitive young consumers and fashion garment repair: Emotional connections to garments as a sustainability strategy. *International Journal of Consumer Studies, 44*(4), 361–368. https://doi.org/10.1111/ijcs.12572

McNeill, L. S., Hamlin, R. P., McQueen, R. H., Degenstein, L., Wakes, S., Garrett, T. C., & Dunn, L. (2020). Waste not want not: Behavioural intentions toward garment life extension practices, the role of damage, brand and cost on textile disposal. *Journal of Cleaner Production, 260*. https://doi.org/10.1016/j.jclepro.2020.121026

Mehta, P., Kaur, A., Singh, S., & Mehta, M. D. (2023). "Sustainable attitude" – a modest notion creating a tremendous difference in the glamourous fast fashion world: Investigating moderating effects. *Society and Business Review, 18*(4), 549–571. https://doi.org/10.1108/SBR-10-2021-0205

Melnyk, V., Carrillat, F. A., & Melnyk, V. (2021). The influence of social norms on consumer behavior: A meta-analysis. *Journal of Marketing, 86*(3), 98–120. https://doi.org/10.1177/00222429211029199

Miller, M. R., & Burbach, M. E. (2020). Cigarette butt disposal behavior: A case study of a public beach on Jekyll Island, Georgia, USA.

Moon, K. K. L., Lai, C. S. Y., Lam, E. Y. N., & Chang, J. M. T. (2015). Popularization of sustainable fashion: Barriers and solutions. *Journal of the Textile Institute, 106*(9), 939–952. https://doi.org/10.1080/00405000.2014.955293

Nanayakkara, N. (2019). Conceptual review on apparel disposal behaviour of consumers. *Sri Lanka Journal of Marketing, 5*(1).

Nigbur, D., Lyons, E., & Uzzell, D. (2010). Attitudes, norms, identity and environmental behaviour: Using an expanded theory of planned behaviour to predict

participation in a kerbside recycling programme. *British Journal of Social Psychology, 49*(2), 259–284. https://doi.org/10.1348/014466609X449395

Nook, E. C., Ong, D. C., Morelli, S. A., Mitchell, J., & Zaki, J. (2016). Prosocial conformity. *Personality and Social Psychology Bulletin, 42*(8), 1045–1062. https://doi.org/10.1177/0146167216649932

Norum, P. S. (2015). Trash, charity, and secondhand stores: An empirical analysis of clothing disposition. *Family and Consumer Sciences Research Journal, 44*(1), 21–36. https://doi.org/10.1111/fcsr.12118

Paço, A. do, Filho, W. L., Ávila, L. V., & Dennis, K. E. (2020). Fostering sustainable consumer behavior regarding clothing: Assessing trends on purchases, recycling and disposal. *Textile Research Journal.* https://doi.org/10.1177/0040517520944524

Paden, N., & Stell, R. (2005). Consumer product redistribution. *Journal of Marketing Channels, 12*(3), 105–123. https://doi.org/10.1300/J049v12n03_06

Park, H.-H., Choo, T.-G., & Ku, Y.-S. (2016). The influence of shopping orientation on difficulty discarding and disposal behavior of fashion products. *Fashion & Textile Research Journal, 18*(6), 833–843. https://doi.org/10.5805/sfti.2016.18.6.833

Pereira, G. A., Resende, M., & Cirillo, M. Â. (2020). Generalized ridge estimators adapted in structural equation models. *Acta Scientiarum - Technology, 43*, 1–10. https://doi.org/10.4025/actascitechnol.v43i1.49929

Pristl, A. C., Kilian, S., & Mann, A. (2021). When does a social norm catch the worm? Disentangling social normative influences on sustainable consumption behaviour. *Journal of Consumer Behaviour, 20*(3), 635–654. https://doi.org/10.1002/cb.1890

Shim, S. (1995). Environmentalism and consumers' clothing disposal patterns: An exploratory study. *Clothing and Textiles Research Journal, 13*(1), 38–48. https://doi.org/10.1177/0887302X9501300105

Silva, S. C. e, Santos, A. M. R., Duarte, P., & Vlačić, B. (2021). The role of social embarrassment, sustainability, familiarity and perception of hygiene in second-hand clothing purchase experience. *International Journal of Retail & Distribution Management, 49*(6), 717–734. https://doi.org/10.1108/IJRDM-09-2020-0356

Sonnenberg, N. C., Marx-Pienaar, J. M. M., & Stols, M. J. (2019). Female consumers' apparel disposal behaviour in the South African emerging market context. *WIT Transactions on Ecology and the Environment, 231*, 281–285. https://doi.org/10.2495/WM180261

Ssemugabo, C., Wafula, S. T., Lubega, G. B., Ndejjo, R., Osuret, J., Halage, A. A., & Musoke, D. (2020). Status of household solid waste management and associated factors in a slum community in Kampala, Uganda. *Journal of Environmental and Public Health, 2020*(1), 6807630. https://doi.org/10.1155/2020/6807630

Sumo, P. D. (2022). The retail second-hand clothing sector in developing economy: Case study of Liberia. *Croatian Regional Development Journal, 3*(2), 139–160. https://doi.org/10.2478/crdj-2022-0013

Sung, K., Cooper, T., & Kettley, S. (2019). Factors influencing upcycling for UK makers. *Sustainability, 11*(3), 870.

Tang, Z., Chen, X., & Luo, J. (2010). Determining socio-psychological drivers for rural household recycling behavior in developing countries. *Environment and Behavior.* https://doi.org/10.1177/0013916510375681

Tonglet, M., Phillips, P. S., & Read, A. D. (2004). Using the theory of planned behaviour to investigate the determinants of recycling behaviour: A case study from Brixworth, UK. *Resources, Conservation and Recycling, 41*. https://doi.org/10.1016/j.resconrec.2003.11.001

Triandis, H. C. (1977). Theoretical framework for evaluation of cross-cultural training effectiveness. *International Journal of Intercultural Relations, 1*(4), 19–45. https://doi.org/10.1016/0147-1767(77)90030-X

Udin, U., Dananjoyo, R., Sugianto, L. O., Ula, H. L., Khairunnisa, R., & Ihsaniati, N. S. N. (2024). Leading by example: Creating the environmental care village based on waste donation. *Multidisciplinary Science Journal, 6*(11). https://doi.org/10.31893/multiscience.2024239

Vlastelica, T., Kostić-Stanković, M., Krstić, J., & Rajić, T. (2023). Generation Z's intentions towards sustainable clothing disposal: Extending the theory of planned behavior. *Polish Journal of Environmental Studies, 32*(3), 2345–2360. https://doi.org/10.15244/pjoes/157007

Wan, C., Shen, G. Q., & Yu, A. (2014). The role of perceived effectiveness of policy measures in predicting recycling behaviour in Hong Kong. *Resources, Conservation and Recycling, 83*, 141–151.

Wang, Z., Guo, D., & Wang, X. (2016). Determinants of residents' e-waste recycling behaviour intentions: Evidence from China. *Journal of Cleaner Production, 137*. https://doi.org/10.1016/j.jclepro.2016.07.155

Weber, S., Lynes, J., & Young, S. B. (2017). Fashion interest as a driver for consumer textile waste management: Reuse, recycle or disposal. *International Journal of Consumer Studies, 41*(2), 207–215. https://doi.org/10.1111/ijcs.12328

White, K., Habib, R., & Hardisty, D. J. (2019). How to SHIFT consumer behaviors to be more sustainable: A literature review and guiding framework. *Journal of Marketing, 83*(3), 22–49. https://doi.org/10.1177/0022242919825649

Yamin, P., Fei, M., Lahlou, S., & Levy, S. (2019). Using social norms to change behavior and increase sustainability in the real world: A systematic review of the literature. *Sustainability, 11*(20), 5847. https://doi.org/10.3390/su11205847

Yang, J., Al Mamun, A., Reza, M. N. H., Yang, M., & Aziz, N. A. (2024). Predicting the significance of consumer environmental values, beliefs, and norms for sustainable fashion behaviors: The case of second-hand clothing. *Asia Pacific Management Review, 29*(2), 179–194. https://doi.org/10.1016/j.apmrv.2024.01.001

Yee, L. W., Hassan, S. H., & Ramayah, T. (2016). Sustainability and philanthropic awareness in clothing disposal behavior among young Malaysian consumers. *Sage Open.* https://doi.org/10.1177/2158244015625327

Zalpuri, R., Sharma, J. K., Singh, R., & Rawat, L. (2022). Disposal practices and awareness of medicine waste management among the general population of Delhi-National Capital Region, India. *Journal of Applied Pharmaceutical Science, 12*(1), 078–083. https://doi.org/10.7324/JAPS.2021.120107

Zhang, Y., Zhang, M., Weng, Z., Gao, X., & Liao, W. (2023). The influence of social norms and environmental regulations on rural households' pesticide packaging waste disposal behavior. *Sustainability, 15*(22), 15938.